DRAM Controllers and Address Mapping

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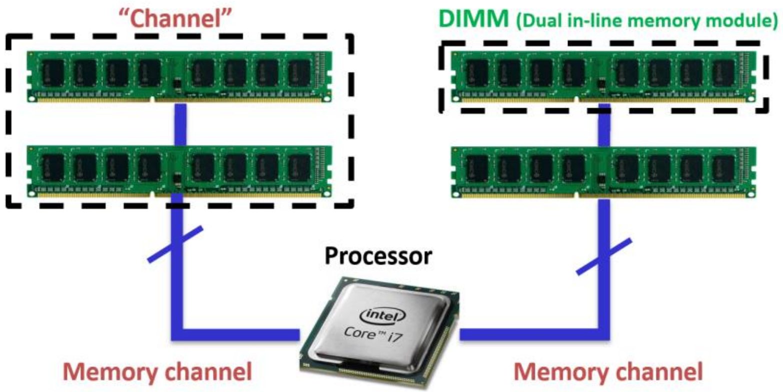
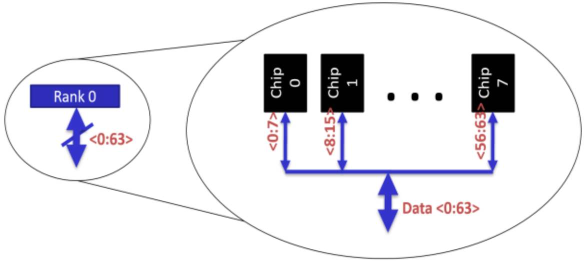
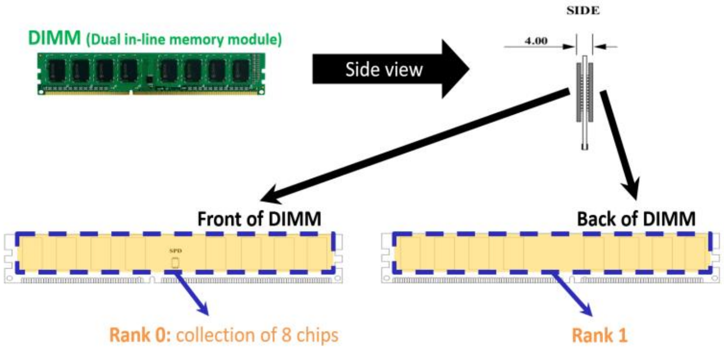
DRAM Subsystem Organization

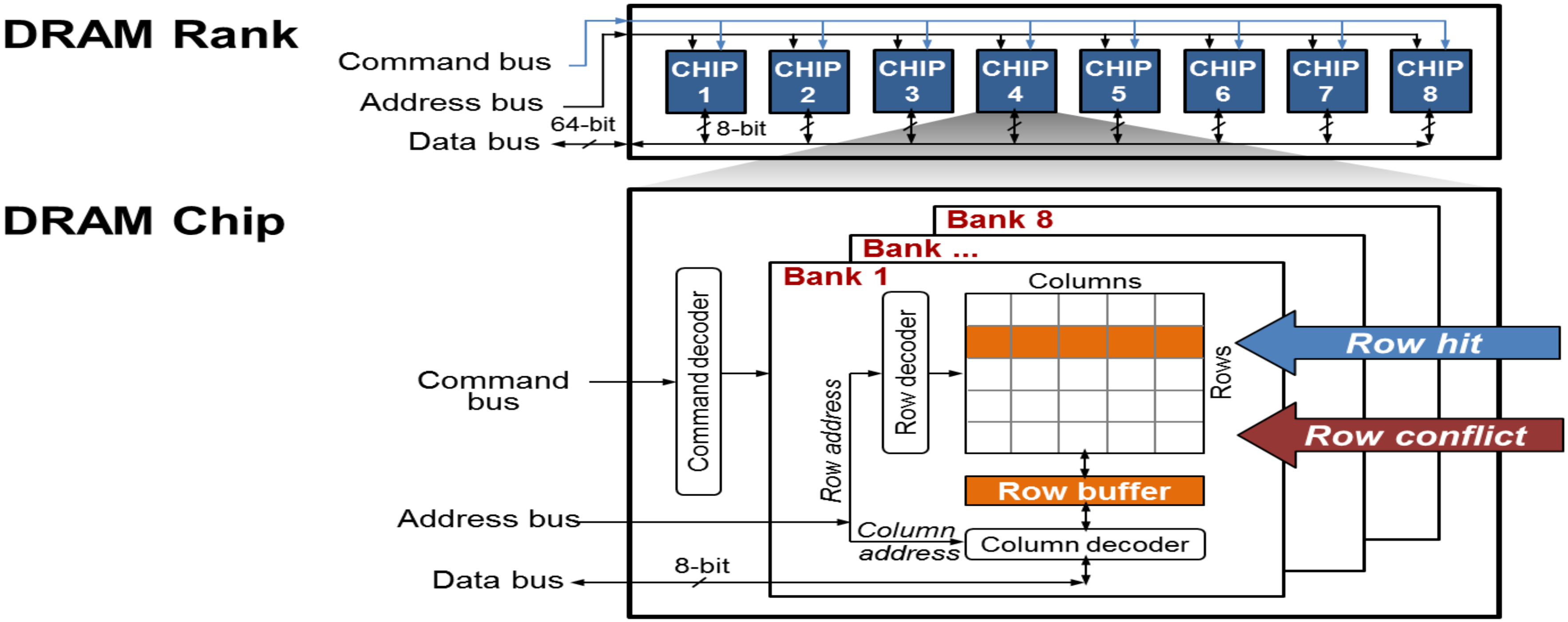
- ❖ Channel
- ❖ DIMM
- ❖ Rank
- ❖ Chip
- ❖ Bank
- ❖ Row
- ❖ Column
- ❖ B-Cell







DRAM Rank



DRAM access latency varies depending on which row is stored in the row buffer

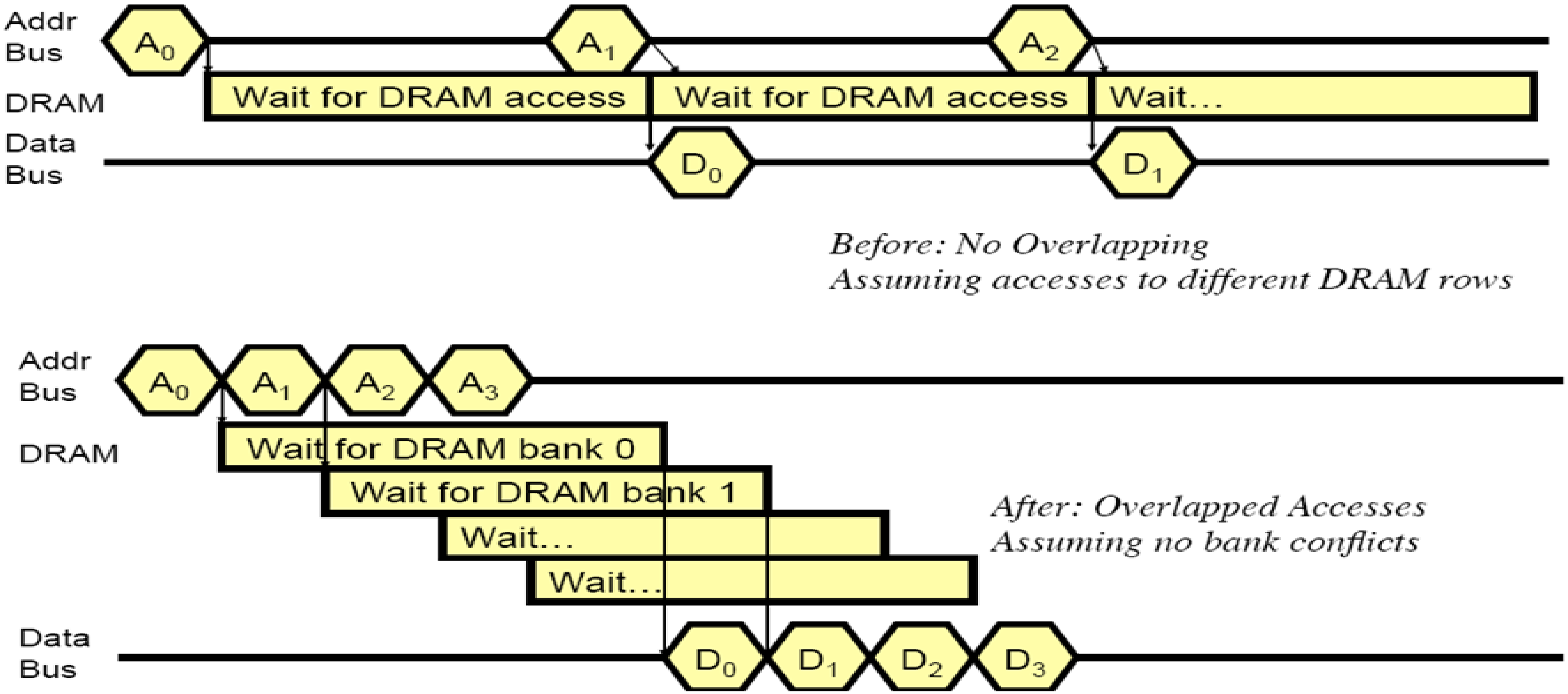
Basic DRAM Operation

- CPU → controller transfer time
- Controller latency
  - Queuing & scheduling delay at the controller
  - Access converted to basic commands
- Controller → DRAM transfer time : Bus latency (BL)
- DRAM bank latency
  - Simple CAS (column address strobe) if row is “open”
  - RAS (row address strobe) + CAS if array precharged
  - PRE + RAS + CAS (worst case)
- DRAM → Controller transfer time : Bus latency (BL)
- Controller to CPU transfer time

Multiple Banks and Channels

- **Multiple banks**
  - Enable concurrent DRAM accesses
  - Bits in address determine which bank an address resides in
- **Multiple independent channels**
  - Fully parallel as they have separate data buses
  - Increased bus bandwidth
  - More wires, area and power consumptions
  - More pins for on-chip memory controller
- Enabling more concurrency requires reducing
  - Bank conflicts, Channel conflicts

Multiple banks to reduce delay



Address Mapping (Single Channel)

- Single-channel system, 8B memory bus
  - 2GB memory, 8 banks, 16K rows & 2K columns per bank
- Row interleaving
  - Consecutive rows of memory in consecutive banks
  - Accesses to consecutive rows serviced in a pipelined manner

Row (14 bits)	Bank (3 bits)	Column (11 bits)	Byte in bus (3 bits)

- Cache block interleaving
  - Consecutive cache block addresses in consecutive banks
  - 64 B cache blocks, Accesses to consecutive cache blocks in parallel

Row (14 bits)	High Column	Bank (3 bits)	Low Col.	Byte in bus (3 bits)
	8 bits		3 bits	

Address Mapping (Multiple Channels)

C	Row (14 bits)	Bank (3 bits)	Column (11 bits)	Byte in bus (3 bits)

Row (14 bits)	C	Bank (3 bits)	Column (11 bits)	Byte in bus (3 bits)

Row (14 bits)	Bank (3 bits)	C	Column (11 bits)	Byte in bus (3 bits)

Row (14 bits)	Bank (3 bits)	Column (11 bits)	C	Byte in bus (3 bits)

- ❖ Where are consecutive cache blocks?

C	Row (14 bits)	High Column 8 bits	Bank (3 bits)	Low Col. 3 bits	Byte in bus (3 bits)

Row (14 bits)	C	High Column 8 bits	Bank (3 bits)	Low Col. 3 bits	Byte in bus (3 bits)

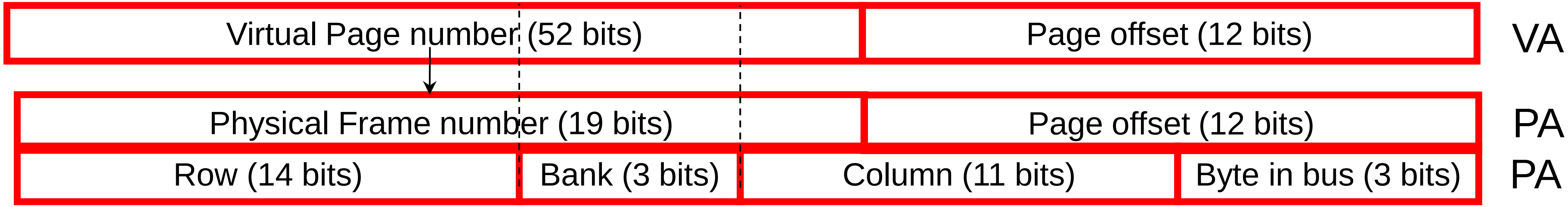
Row (14 bits)	High Column 8 bits	C	Bank (3 bits)	Low Col. 3 bits	Byte in bus (3 bits)

Row (14 bits)	High Column 8 bits	Bank (3 bits)	C	Low Col. 3 bits	Byte in bus (3 bits)

Row (14 bits)	High Column 8 bits	Bank (3 bits)	Low Col. 3 bits	C	Byte in bus (3 bits)

Virtual To Physical Address Mapping

- ❖ Operating System influences where an address maps to in DRAM



- ❖ Operating system can influence which bank/channel/rank a virtual page is mapped to.

- ❖ It can perform page coloring to
  - ❖ Minimize bank conflicts
  - ❖ Minimize inter-application interference

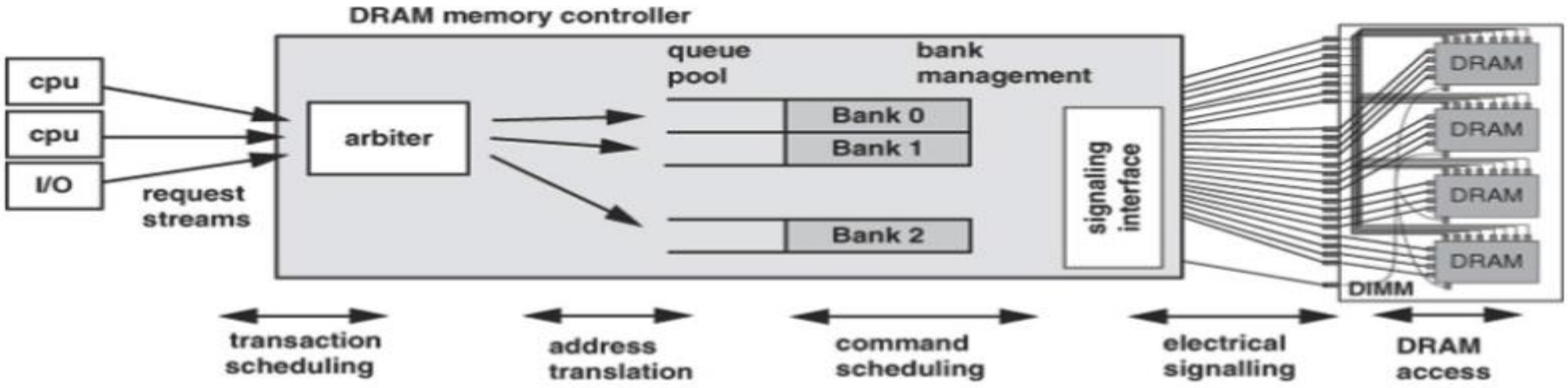
DRAM Controller: Functions

- ❖ Ensure correct operation of DRAM
  - ❖ Address mapping, refresh and timing
- ❖ Service DRAM requests while obeying timing constraints of DRAM chips
  - ❖ Constraints: resource conflicts (bank, bus, channel), minimum write-to-read delays
- ❖ Translate requests to DRAM command sequences
- ❖ Buffer and schedule requests to improve performance
  - ❖ Reordering, row-buffer, bank, rank, bus management
- ❖ Manage power consumption and thermals in DRAM
  - ❖ Turn on/off DRAM chips, manage power modes

DRAM Controller: Where to Place ?

- **In chipset**
- More flexibility to plug different DRAM types into the system
- Less power density in the CPU chip
- **On CPU chip**
- Reduced latency for main memory access
- Higher bandwidth between cores and controller
- More information can be communicated between CPU and controller

DRAM Controller Overview



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